Correct° Exam SL (Tekup) [21/22]

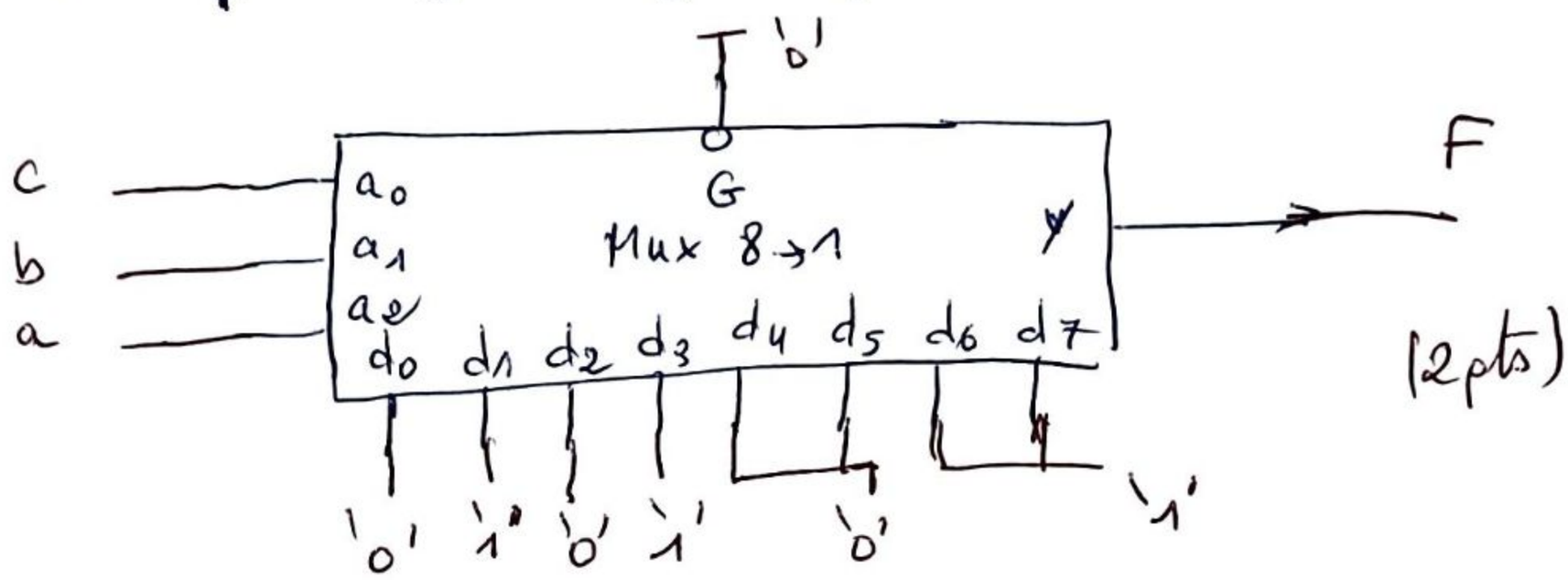
Exercice 1 : (6 pts)

$F = a.b + \bar{a}.c$

1) $F(a,b,c) = a.b.(c+\bar{c}) + \bar{a}.(b+\bar{b}).c$

$= a.b.c + a.b\bar{c} + \bar{a}.b.c + \bar{a}.\bar{b}.c$ (1pt)

$F(a,b,c) = m_7 + m_6 + m_3 + m_1$



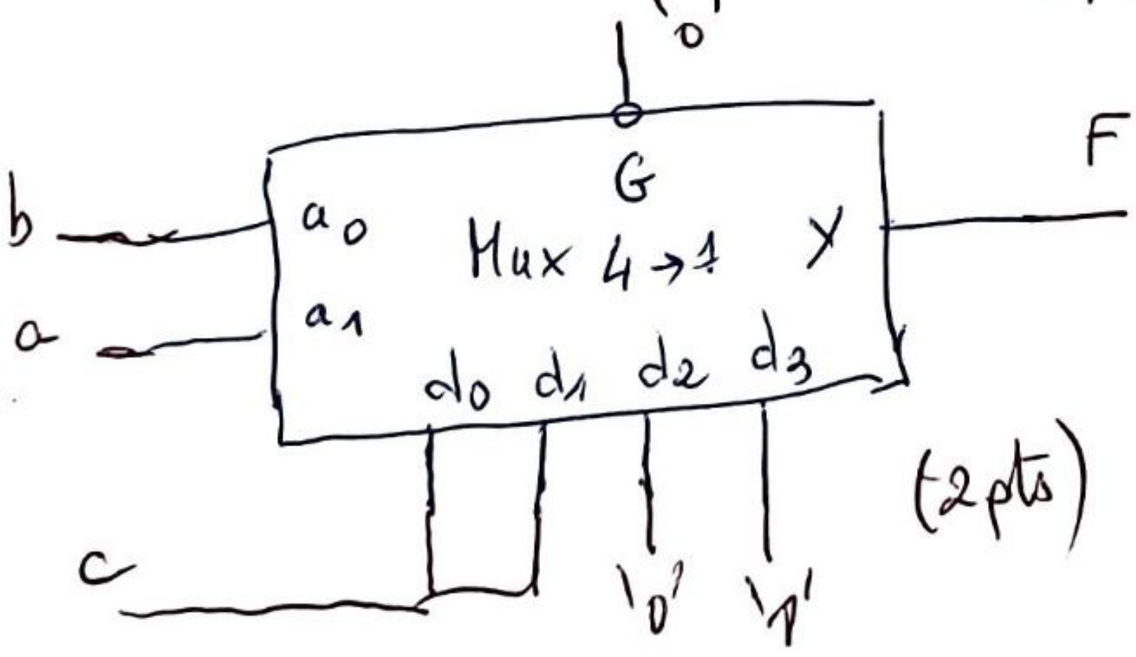
(2pts)

2) $F(a,b,c) = a.b + \bar{a}.c$

$= m_3 + \bar{a}.(\bar{b}+b).c$

$= m_3 + \bar{a}.\bar{b}.c + \bar{a}.b.c$

$= m_3 + m_0.c + m_1.c$ (1pt)



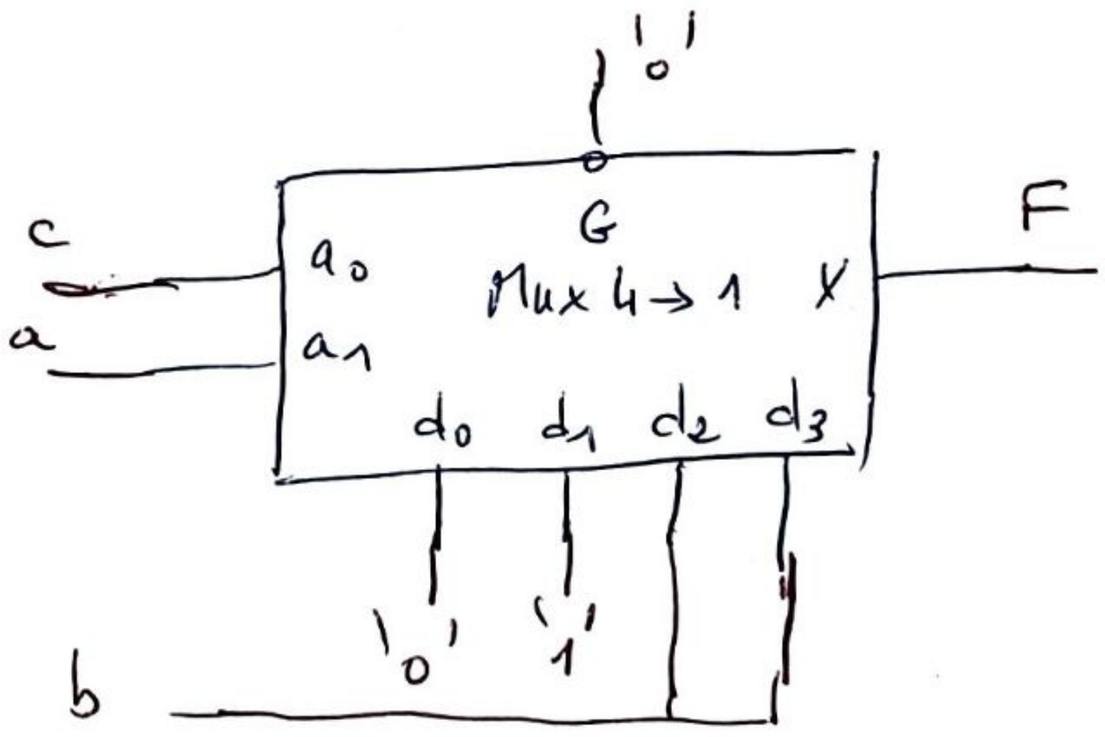
(2pts)

$F(a,b,c) = a.b + \bar{a}.c$

$= a.(\bar{c}+c).b + m_1$

$= m_1 + a.\bar{c}.b + a.c.b$

$= m_1 + m_2.b + m_3.b$



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Exercice 2 (6 pts)

1) $F(A,B,C,D) = m_0 \cdot d_0 + m_1 \cdot d_1 + m_2 \cdot d_2 + m_3 \cdot d_3 + m_4 \cdot d_4 + m_5 \cdot d_5 + m_6 \cdot d_6 + m_7 \cdot d_7$ (2pts)

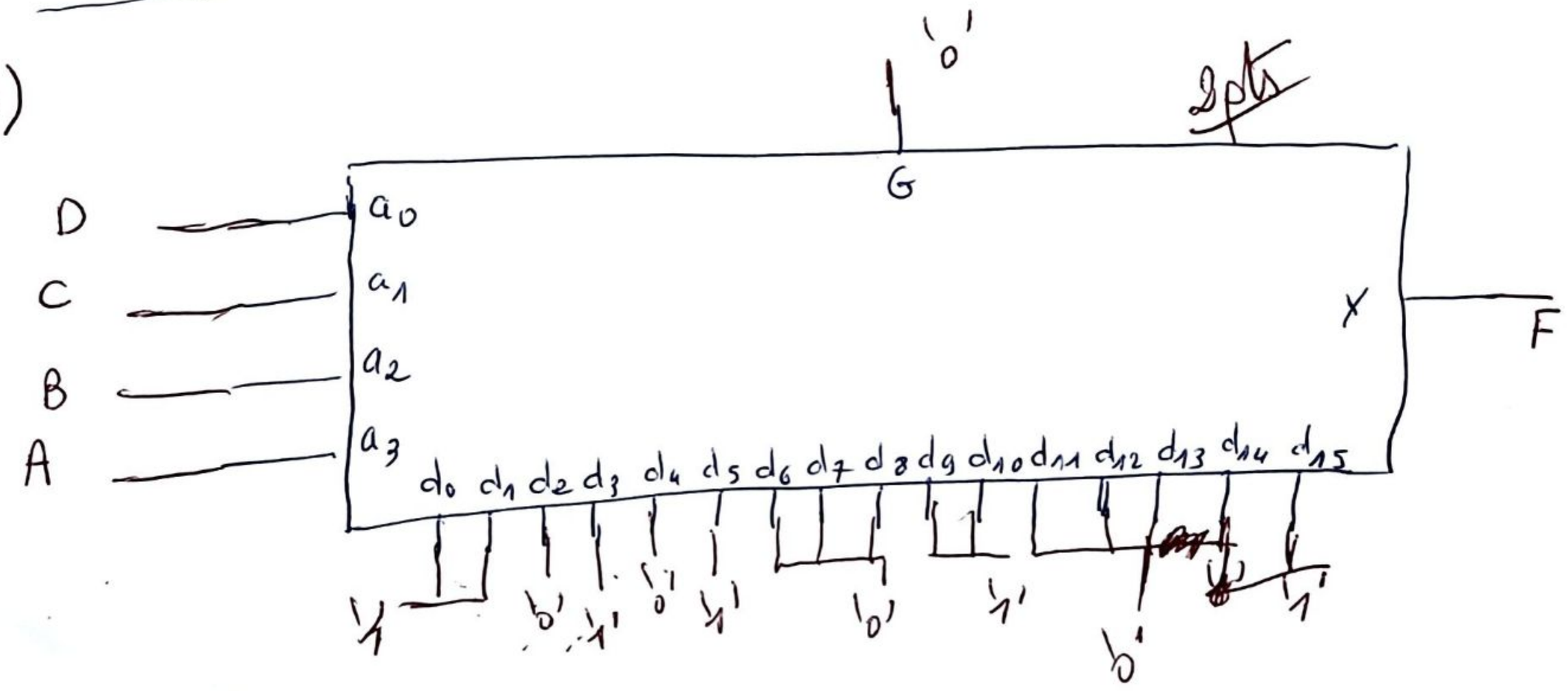
$= \underbrace{\bar{A}.\bar{B}.\bar{C}.\bar{D}}_{0} + \underbrace{\bar{A}.\bar{B}.C.D}_{3} + \underbrace{\bar{A}.\bar{B}.C.D}_{5} + \underbrace{\bar{A}.B.C.\bar{D}}_{9} + \underbrace{A.\bar{B}.\bar{C}.D}_{10} + \underbrace{A.\bar{B}.C.D}_{15} + \underbrace{A.B.C.\bar{D}}_{14}$ (2pts)

2)

A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

(2 pts)

3)



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Exercice 3 (8 pts)

1/ $J_0 = \overline{Q_2}$ 0,5 pt

$K_0 = Q_2$ 0,5 pt

$J_1 = Q_0$ 0,5 pt

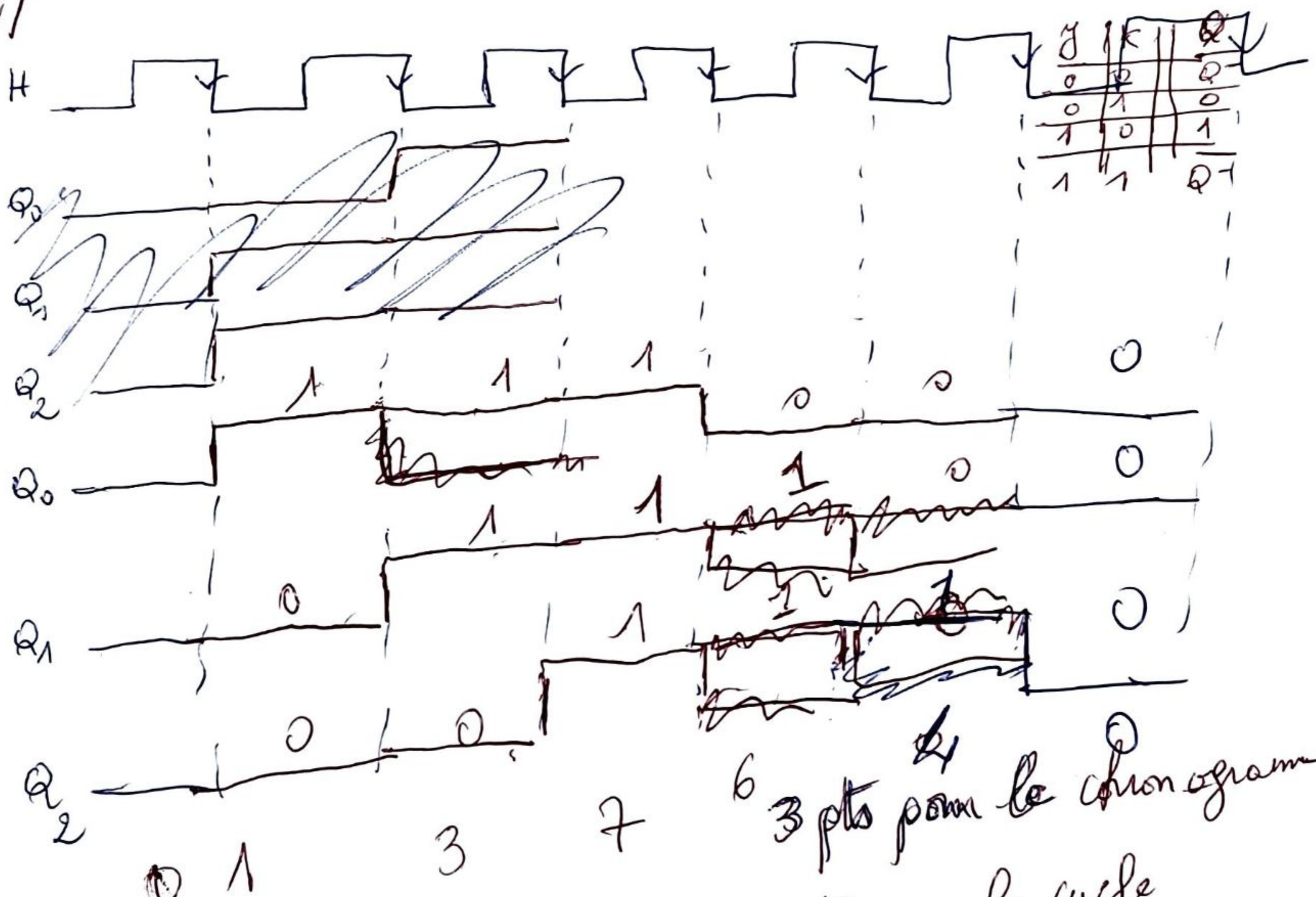
$K_1 = \overline{Q_0}$ 0,5 pt

$J_2 = \overline{\overline{Q_0 \cdot Q_1}} = Q_0 \cdot Q_1$ 0,5 pt

$K_2 = \overline{Q_1}$ 0,5 pt

1	1	0	0	0	0	0	0
0	0	1	1	1	1	1	1
0	1	1	1	0	0	0	0
1	1	0	0	1	1	1	1
0	0	1	1	0	0	0	0
1	0	0	0	1	0	1	1

2/



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